

DANY DIAB

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Halifax, NS | Willing to relocate for Summer 2027

PROFILE

Graphics and engine developer, developing a CUDA-accelerated path tracer with a Vulkan backend and built a C++ / OpenGL L-system parser and renderer using GPU instancing. Seeking internship in graphics and game engine development.

PROJECTS

GPU Path Tracer | C++, CUDA (*in progress*)

- Developing a physically-based path tracer in C++ with CUDA for GPU-accelerated rendering.
- Implemented a Monte Carlo path tracing integrator with physically based BSDF allowing for diffuse, specular, and dielectric materials.
- Currently adding Vulkan backend support and BVH acceleration for scene complexity scaling.

L-System Parser and Renderer | C++, OpenGL

- Built a custom parser and renderer for parametric and stochastic L-system production rules, supporting both 2D and 3D rendering.
- Implemented GPU instancing shaders to render cylinders efficiently at scale, enabling complex organic structures to emerge from rule evaluation.
- Developed quaternion logic for turtle graphics and geometry orientation for accurate 3D rotations.

Procedural Terrain Sandbox | C#, Unity

- Engineered a procedural terrain generation system implementing Fractal Brownian Motion (fBm) and Ridge Noise algorithms from scratch to simulate realistic mountain topography.
- Generated 3D meshes from noise data using Unity's Burst Compiler and Job System for multi-threaded performance.
- Implemented terrain save/load using JSON serialization and Unity ScriptableObjects, enabling users to export and share generated terrain via a custom file.

EXPERIENCE

Research Assistant, Simulation Development | Dalhousie University, GEM Lab | Summer 2026

- Built a simulation environment integrating real-world bathymetry and backscatter data from the Bay of Fundy and Gulf of St. Lawrence.
- Led development of core simulation systems including physics tuning, object spawning, and collision setup.
- Implemented trawl line systems, lobster trap spawning, and colliders for the simulated fishing environment.

Internship, WERFY | Montreal, QC | Summer 2025

- Built a data migration utility in Python and PowerShell to extract, classify, and reorganize unstructured company file system data using NLP scripts.
- Developed and deployed internal automation tools, including a bulk email drafting system and an attachment extraction pipeline.
- Integrated external APIs with a cloud platform to automatically generate and extract performance reports and key business insights.

EDUCATION

Bachelor of Computer Science
Dalhousie University | 2023 - 2027 | GPA: 3.5

SKILLS

C++, Unity, CUDA, OpenGL, HLSL, Vulkan (in progress), C#, C, Python, Git